

Preliminary validation of RELAP5/Mod4.0 code for LBE cooled NACIE facility



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HIGHLIGHTS

- Detail discussion of thermo physical properties of Lead Bismuth Eutectic incorporated in the code RELAP5/Mod4.0 included.
- Benchmarking of LBE properties in RELAP5/Mod4.0 against literature.
- NACIE facility for three different power levels (10.8, 21.7 and 32.5 kW) under natural circulation considered for benchmarking.
- Preliminary validation of the LBE properties against experimental data.
- NACIE facility for power level 22.5 kW considered for validation.

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ABSTRACT

The one-dimensional thermal hydraulic computer code RELAP5 was developed for thermal hydraulic study of light water reactor as well as for nuclear research reactors. The purpose of this work is to evaluate the code RELAP5/Mod4.0 for analysis of research reactors. This paper consists of three major sections. The first section presents detailed discussions on thermo-physical properties of Lead Bismuth Eutectic (LBE) incorporated in RELAP5/Mod4.0 code. In the second section, benchmarking of RELAP5/Mod4.0 has been done with the Natural Circulation Experimental (NACIE) facility in comparison with Barone's simulations using RELAP5/Mod3.3. Three different power levels (10.8 kW, 21.7 kW and 32.5 kW) under natural circulation conditions are considered. Results obtained for LBE temperatures, temperature difference across heat section, pin surface temperatures, mass flow rates and heat transfer coefficients in heat section heat exchanger are in agreement with Barone's simulation results within 7% of average relative error. Third section presents validation of RELAP5/Mod4.0 against the experimental data of NACIE facility performed by Tarantino et al. test number 21 at power of 22.5 kW comparing the profiles of temperatures, mass flow rate and velocity of LBE. Simulation and experimental results agree within 7% of average relative error.

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1. Introduction

RELAP5 one-dimensional thermal-hydraulic computer code was developed for analysis of transients as well as normal operation of light water reactors. The RELAP5 code includes models of several hydrodynamic components (pipes, valves, pumps, separators etc.) using which different loops can be simulated (Roux, 2001; Talukder and Bhuyan, 2012). RELAP5 solves six (three for each phase) basic equations of energy, mass, and momentum. There are eight primary independent variables in these equations: void fraction, phasic velocities, pressure, phasic specific internal energies, boron density and noncondensable quality. Six secondary

dependent variables included in these equations are: phasic temperature, phasic densities, saturation temperature and noncondensable mass fraction. Two independent variables are distance and time (Roux, 2001). This code can also be used to simulate the pressure of non-condensable gases. Models of interphase mass transfer, interphase drag, wall friction and flow regimes are also included in RELAP5 (Talukder and Bhuyan, 2012; Licht and Dionne, 2015).

Lead-Bismuth Eutectic (LBE) is a mixture of lead (44.5 wt.%) and bismuth (55.5 wt.%). It is proposed to be used in most accelerator driven system (ADS) designs. LBE as a spallation target material produces high quantity neutron in spallation reaction; has small scattering and neutron absorption cross-section. LBE has good characteristics: high boiling point (~1670 °C) which facilitates the reactor to be operated at high temperatures; chemical

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Nomenclature

HEX	heat exchanger
LBE	Lead Bismuth Eutectic
HS	heat section

NAT-1	natural circulation test-1
NAT-2	natural circulation test-2
NAT-3	natural circulation test-3

inertness towards water and air prevents the use of intermediate coolant giving low capital cost; low melting temperature; good thermal, chemical and safety characteristics. However, LBE has some disadvantages: more corrosive to steel than sodium; material cost is high; the radioactive polonium Po-210 generated by neutron capture of Bi-209 (Srinu, 2010; Koji et al., 2006; Handbook on Lead-Bismuth Eutectic, 2007, 2015; IAEA Nuclear Energy Series No. NP-T-1.6, 2012; Amaya et al., 2009).

1.1. Thermodynamic properties of LBE

The one dimensional code RELAP5/Mod4.0 working for water-steam system was modified by incorporating the properties of LBE to perform the simulation using liquid metal (Srinu, 2010). Following thermo physical properties (for pressure up to ~0.1 MPa) (Handbook on Lead-Bismuth Eutectic, 2007) of the Lead Bismuth Eutectic (LBE) are incorporated in the RELAP5/Mod4.0 code:

1.1.1. Density

Density as the function of temperature is useful in developing equation of state (EOS). It is used for the calculation of thermal diffusivity, viscosity, surface tension and concentration per unit volume. Density of the liquid decreases with the increase of temperature. Density of LBE as the function of temperature (403–1100 K) is given by Handbook on Lead-Bismuth Eutectic (2007):

$$\rho_{LBE} [kg\ m^{-3}] = 11096 - 1.3236 \times T \quad (1)$$

where T is temperature in Kelvin.

1.1.2. Saturation vapour pressure

The following correlation is recommended for the saturation vapour pressure of LBE in terms of temperature (508–1943 K):

$$p_{LBE} [Pa] = 1.11 \times 10^{10} \cdot \exp\left(-\frac{22552}{T}\right) \quad (2)$$

1.1.3. Normal melting point

The recommended normal melting point of LBE at atmospheric pressure is given by:

$$T_{meltLBE} [K] = 397.7 \pm 0.6 \quad (3)$$

1.1.4. Normal boiling point

The recommended normal boiling point of LBE at atmospheric pressure is given by:

$$T_{boilLBE} [K] = 1943 \pm 50 \quad (4)$$

1.1.5. Critical constants

Following data of critical temperature, pressure and density for LBE were used in the RELAP5/Mod4:

$$T_{cLBE} [K] = 4890 \quad (5)$$

$$P_{cLBE} [MPa] = 88 \quad (6)$$

$$\rho_{cLBE} [kg/m^3] = 2170 \quad (7)$$

1.1.6. Heat capacity

The following correlation were used in RELAP5/Mod4.0 for the calculation of heat capacity of LBE in the temperature range 430–605 K:

$$C_{pLBE} [J\ kg^{-1}K^{-1}] = 159 - 2.72 \times 10^{-2} \times T + 7.12 \times 10^{-6} \times T^2 \quad (8)$$

1.1.7. Heat of vaporization

The following data of heat of vaporization at normal boiling point for LBE were used in the RELAP5:

$$Q_{boilLBE} = 178.0 \pm 0.074\ kJ/mole = 854 \pm 2\ kJ/kg \quad (9)$$

1.1.8. Latent heat of melting

Similarly latent heat of melting at atmospheric pressure is given by:

$$Q_{meltLBE} = 8.01 \pm 0.07\ kJ/mole = 38.5 \pm 0.3\ kJ/kg \quad (10)$$

1.1.9. Enthalpy, entropy and internal energy

The recommended correlations for the calculation of enthalpy, entropy and internal energy of LBE (for 44.29 at.%Pb and 55.71 at.% Bi and purity 99.99% at 400 K) were inserted in the RELAP5/Mod4.0 code as the function of temperature.

$$\Delta H_{LBE}^{298} [T] = -5.296 \times 10^{-4} \times T^2 + 29.5182 \times T - 3513.20 \quad (11)$$

$$S(T) = 159 \times \ln\left(\frac{T}{T_m}\right) + 2.72 \times 10^{-2} (T - T_m) + 3.56 \times 10^{-6} (T^2 - T_m^2) \quad (12)$$

$$U(T) = 159 \times (T - T_m) + 1.36 \times 10^{-2} (T^2 - T_m^2) + 2.37 \times 10^{-6} (T^3 - T_m^3) \quad (13)$$

where T_m is the melting point of LBE.

1.1.10. Coefficient of thermal expansion

Density of liquid decreases with the increase of temperature because of thermal expansion of the interatomic forces. The relation between the coefficient of thermal expansion and density is defined as:

$$\beta_x(T) = \frac{1}{V} \cdot \left(\frac{\partial V}{\partial T}\right)_x = -\frac{1}{\rho} \cdot \left(\frac{\partial \rho}{\partial T}\right)_x \quad (14)$$

The coefficient of thermal expansion of molten LBE is given by the following correlation:

$$\beta_{p,LBE} [K^{-1}] = \frac{1}{(8383.2 - T)} \quad (15)$$

1.1.11. Sound velocity and compressibility

Direct measurement data are not available for compressibility of molten LBE. Therefore, adiabatic compressibility is calculated from the sound velocity and density using the following correlation:

$$B_s = -V \left(\frac{\partial p}{\partial V} \right) \bigg|_s \equiv \frac{1}{K_s} = \frac{\rho}{\left(\frac{\partial \rho}{\partial V} \right) \bigg|_s} = \rho \cdot u_{\text{sound}}^2 \quad (16)$$

where ρ is the density and u_{sound} is the sound velocity.

The sound velocity (403–1300 K) and adiabatic compressibility (430–605 K) of LBE is recommended in terms of the temperature:

$$u_{\text{Sound LBE}} [\text{ms}^{-1}] = 1773 + 0.1049 \times T - 2.873 \times 10^{-4} \times T^2 \quad (17)$$

$$B_{\text{SLBE}} [\text{Pa}] = (35.18 + 1.541 \times 10^{-3} \times T - 9.191 \times 10^{-6} \times T^2) \times 10^9 \quad (18)$$

1.1.12. Viscosity

Viscosity (400–1100 K) of LBE is given by:

$$\mu_{\text{LBE}} [\text{Pa.s}] = 4.94 \times 10^{-4} \times \exp \left(\frac{754.1}{T} \right) \quad (19)$$

1.1.13. Surface tension

The recommended correlation for surface tension of molten LBE is given by:

$$\sigma_{\text{LBE}} [\text{Nm}^{-1}] = 0.437 - 6.6 \times 10^{-5} \cdot T \quad (20)$$

where temperature (T) is in Kelvin. This correlation can be used in the temperature range 423–1400 K at normal pressure.

1.1.14. Thermal conductivity and thermal diffusivity

The suggested correlation for the calculation of thermal conductivity (403–1100 K) and thermal diffusivity for LBE is given as follows:

$$\lambda [\text{Wm}^{-1}\text{K}^{-1}] = 3.61 + 1.517 \times 10^{-2} \times T - 1.741 \times 10^{-6} \times T^2 \quad (21)$$

$$\alpha_{\text{PLBE}} [\text{ms}^{-2}] = (1.408 + 0.0112 \times T) \times 10^{-6} \quad (22)$$

1.1.15. Convective heat transfer correlations

The correlation of convective heat transfer coefficient for LBE used in the RELAP5 code is for $40 \leq \text{Pe} \leq 1150$ (Pfrang and Struwe, 2007):

$$\text{Nu} = 5 + 0.025(\text{Pe})^{0.8} \quad (23)$$

where Nu is the Nusselt number, Pe is the Peclet number ($\text{Re} \cdot \text{Pr}$), Re is the Reynolds number and Pr is the Prandtl number.

The properties of LBE incorporated in the modified code RELAP5/Mod4.0 have been benchmarked and validated in the present study using Natural Circulation Experimental (NACIE) facility under the natural circulation condition. RELAP5/Mod4.0 has limitation that it works only for liquid LBE and does not work for two phase of LBE because vapour phase properties of LBE have not been incorporated in the code. Therefore, benchmarking and validation for the forced (gas enhanced) circulation has not been performed.

2. The NACIE facility

The Natural Circulation Experimental (NACIE) facility shown in Fig. 1 designed at ENEA-Brasimone research centre is being used for the thermal hydraulic study, fluid dynamics and control of the nuclear system. The NACIE facility cooling loop is rectangular in shape of 7.5 m height and 1 m width consisting of two vertical and two horizontal stainless steel (AISI 304) pipes (O.D. 6.35 cm).

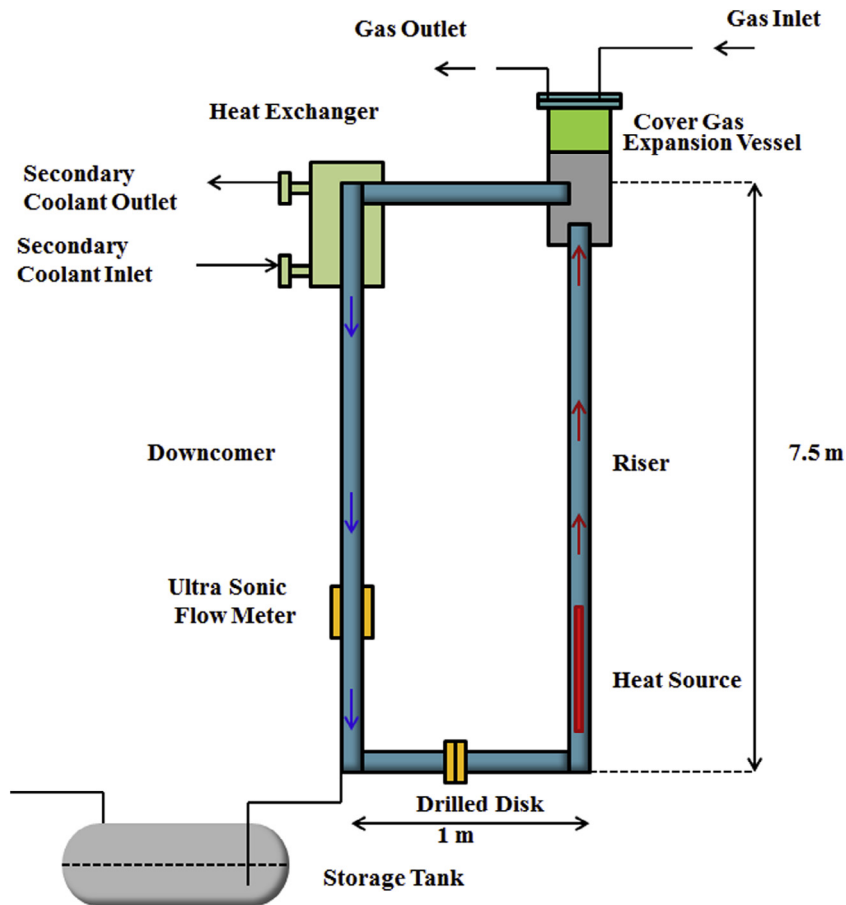


Fig. 1. NACIE facility (adapted from Barone, 2012).

Table 1
Main geometrical data of NACIE facility.

Total vertical length	9231 mm
Horizontal length	1000 mm
Pipe inside diameter	62.7 mm
Pipe thickness	5.16 mm
Expansion vessel height	765 mm
Expansion vessel inside diameter	254.5 mm
Heat exchanger length	1500 mm
Heat source active length	1000 mm

The vertical pipes act as riser and downcomer. Both riser and downcomer are connected to each other by means of horizontal stainless steel pipes (O.D. 6.35 cm). The inventory of LBE in the loop is about 1000 kg. The design temperature and pressure of the loop is about 550 °C and 10 bar respectively. The NACIE facility consists of two loops: LBE mixture in the primary and water in the secondary loops (Barone, 2012; Coccoluto et al., 2011; Tarantino, 2010; Tarantino et al., 2008, 2010, 2011) LBE flows in the tubes and water flows counter-currently in the shell side of the heat

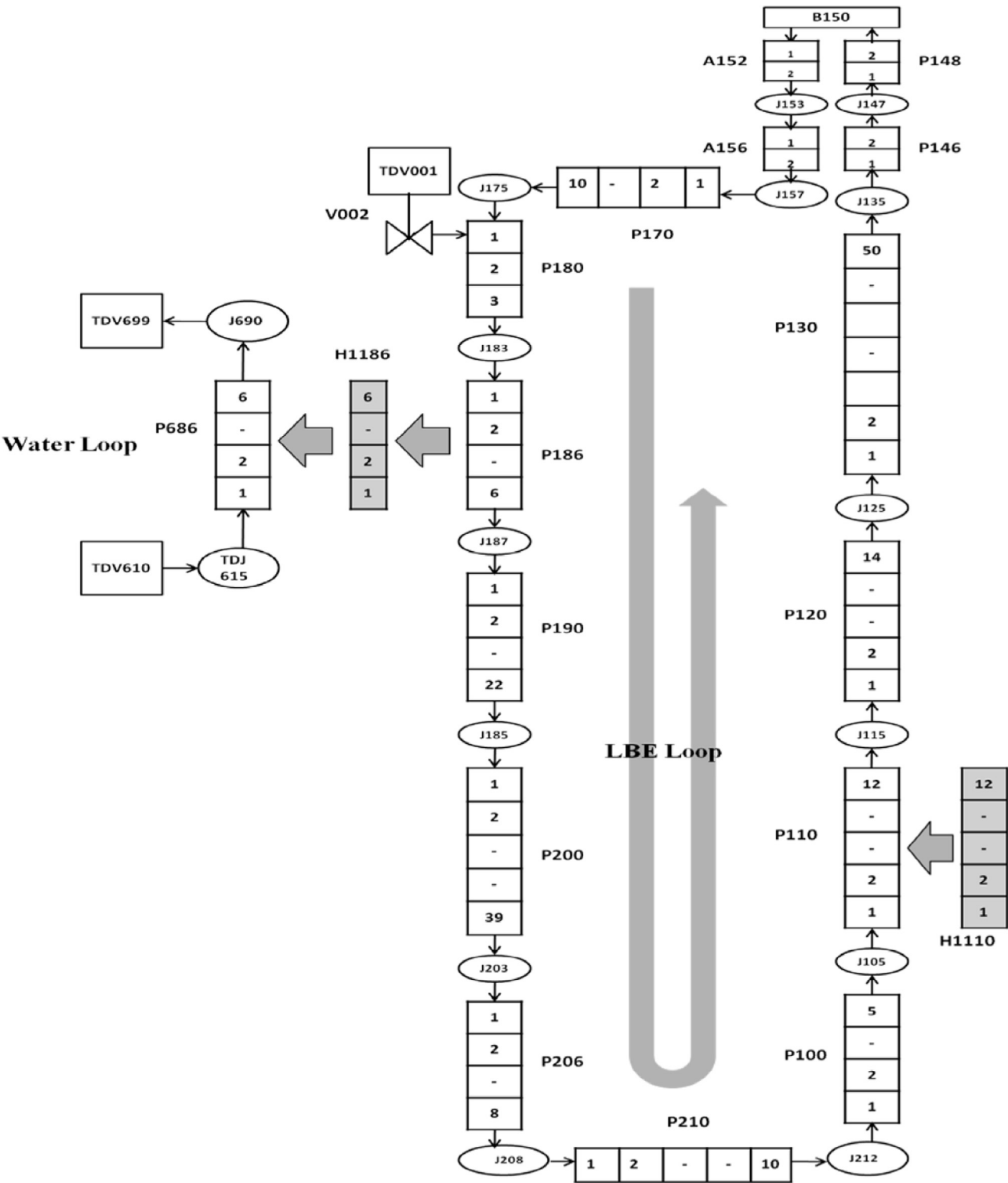


Fig. 2. NACIE facility nodalization (adapted from Barone, 2012).

exchanger (HEX). The heat exchanger (HEX) is connected to upper part of the downcomer. A heat source is installed at lower part of the riser. An expansion vessel is also installed on the top, coaxially to the riser. The main geometrical data of the NACIE facility is reported in Table 1. Both natural circulation and forced circulation tests can be performed using NACIE facility. LBE flows in the natural circulation test due to the driving force results from the thermal buoyancy. However, forced circulation tests are carried out using gas injection technique connected to the riser at the upper part of the heat source. Gas is injected in the riser through a nozzle to circulate LBE, the driving force results from void buoyancy. Finally the gas is separated in the expansion vessel (Barone, 2012; Coccoluto et al., 2011; Tarantino, 2010; Tarantino et al., 2008, 2010, 2011).

3. Benchmarking of RELAP5/Mod4.0

Benchmarking of RELAP5/Mod4.0 code has been carried out against the simulated Barone's NACIE facility. Three case studies NAT-1 (for 10.8 kW), NAT-2 (for 21.7 kW) and NAT-3 (for 32.5 kW) reported in the literature (Barone, 2012) have been benchmarked in this paper. Heat deposited in the LBE by electric pins is removed by heat exchanger. LBE gets heated by the electric pins placed at the lower part of the riser; ascends through the riser into the expansion vessel; enters into the heat exchanger placed at the upper part of the downcomer; flowing through downcomer it reaches the lower part of the heat source completing the cycle. Present work describes benchmarking of RELAP5/Mod4.0 code with the existing literature (Barone, 2012) under the natural condition. Forced circulation using gas injection has not been simulated in this paper. The literature (Barone, 2012) considered for the benchmarking describes the simulations of Natural Circulation Experiment (NACIE) facility using RELAP5/Mod3.3 code.

3.1. RELAP5 nodalization details

Fig. 2 represents RELAP5 nodalization diagram of the NACIE facility (Barone, 2012). Nodalization of the system consists of several pipes, a heat exchanger, an electric heater, various junctions

and boundary conditions. Time dependent volumes TDV610 and TDV699 work as a source and sink for water linked with heat exchanger secondary side (HEX). TDV610 is used to impose water temperature at the inlet, TDV699 is used to impose pressure at the outlet point in the water loop. Time dependent junction TDJ615 is used to circulate water at required flow rate. Time dependent volume TDV001 is used for the initialization of LBE in the closed loop. Heat structures H1110 and H1186 are used to provide the heat boundary condition at the heat source and heat exchanger respectively.

3.2. Initial and boundary conditions

3.2.1. Initial conditions

The initial temperature and pressure of LBE in the primary loop are 563.15 K and 1.0×10^5 Pa respectively (Barone, 2012). The initial temperature and pressure of water in the secondary loop are 393.15 K and 1.6×10^6 Pa respectively (Barone, 2012). Water is circulated at the rate of 2.78 kg/s in the heat exchanger of the secondary loop to extract heat from the primary loop. Wall roughness 3.2×10^{-5} m is taken as suggested in the literature (Barone, 2012; Coccoluto et al., 2011; Tarantino et al., 2008). Heat transfer coefficient of water is taken as a constant value of 4600 W/m²K (Barone, 2012).

3.2.2. Boundary conditions for the heat structures

3.2.2.1. Heat section. Heat section is specifically designed to house 19 fuel pins arranged in hexagonal wrap. Wire spacers are used in place of spacer grids to support the pin bundle. Fig. 3 represents the electrical wire-spaced pin bundle cross section and electrical heater modeling. The dimensions of the bundle are reported in Table 2. Total pin length is 2000 mm and the active length is 600 mm. Wire spacers are installed across the pin length of 1300 mm. The pin bundle has been simulated by the heat structure H1110 embedded inside the hydrodynamic component P110.

3.2.2.2. Heat exchanger (HEX). Shell and tube type heat exchanger is used to transfer heat from LBE to water. Heat exchanger consists of seven tubes whose dimensions are reported in Table 3. The cross

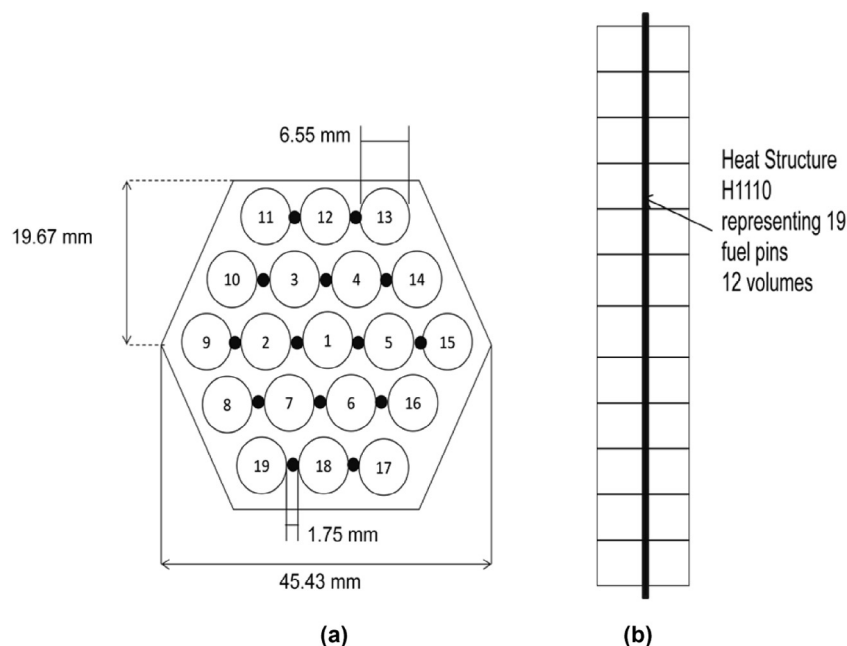


Fig. 3. (a) Electrical wire-spaced pin bundle cross section (adapted from Barone, 2012) (b) Electrical heater modeling.

Table 2
Heat section details.

Rod diameter (d)	6.55 mm
Pitch to diameter ratio (p/d)	1.276
Wire diameter (d_w)	1.75 mm
Wire pitch (H_w)	262 mm

Table 3
Heat exchanger details.

	d_e (mm)	d_i (mm)	t (mm)	Material
Shell (HEX)	406.4	381.0	12.7	AISI 316L
Inner tube (HEX)	73.0	62.7	5.2	AISI 316L
Outer tube (HEX)	88.9	77.9	5.5	AISI 316L

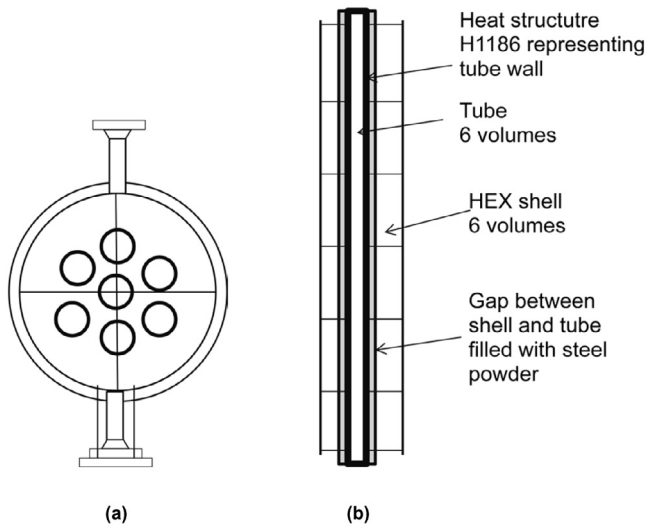


Fig. 4. (a) HEX cross section (adapted from Barone, 2012) (b) HEX heat structure modeling.

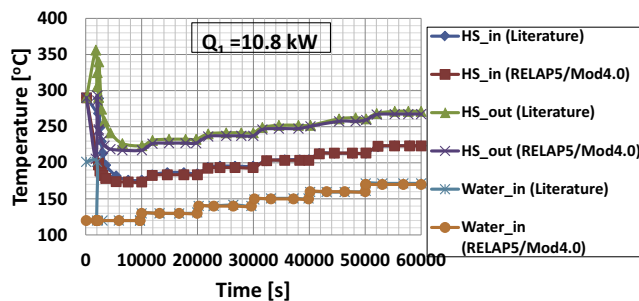


Fig. 5. Comparison of Inlet and outlet LBE temperatures along HS and trend of HEX water inlet temperature (Test NAT-1).

section of HEX and the heat structure used to model the wall of the tube are shown in Fig. 4. Tubes of the heat exchanger are of double-wall type arranged in a hexagonal geometry. The gap between inner and outer tube is filled with the steel powder of thermal conductivity 3 W/m K (Barone, 2012). It is a counter current type of heat exchanger in which LBE flows in downward direction into the tubes, while water flows in the upward direction into the shell side. In order to model the HEX the pipe P186 and the corresponding heat structure are divided into 6 volumes of cell length 0.05 m. The volume flow area of the pipe is equal to the cross flow area of seven tubes of inside diameter 0.0627 m. The volume flow area of

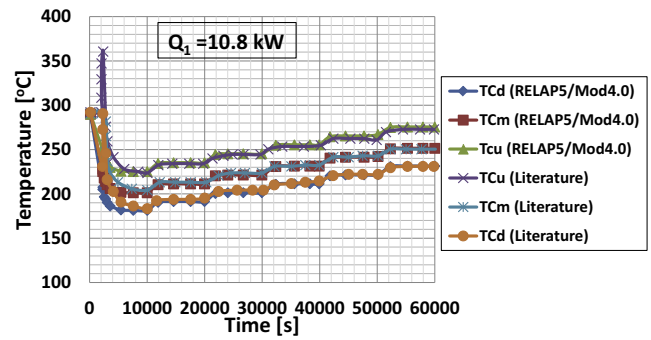


Fig. 6. Comparison of pins surface temperatures (Test NAT-1).

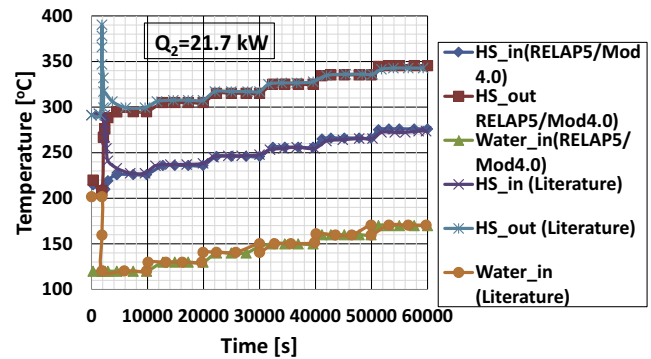


Fig. 7. Comparison of inlet and outlet LBE temperatures along HS and trend of HEX water inlet temperature (Test NAT-2).

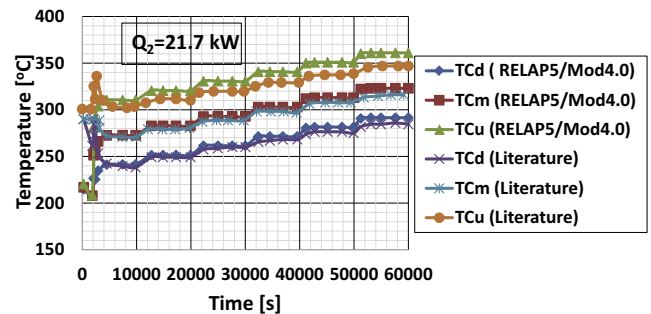


Fig. 8. Comparison of pins surface temperatures (Test NAT-2).

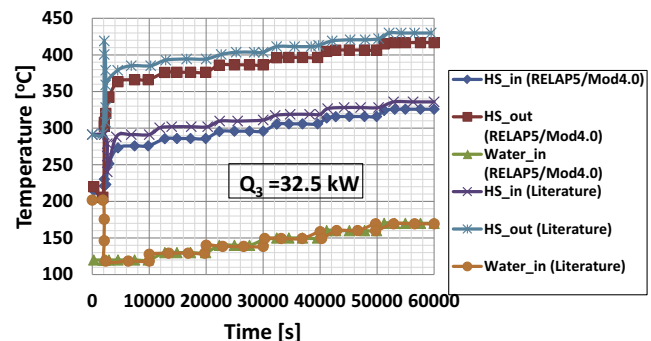


Fig. 9. Comparison of Inlet and outlet LBE temperatures along HS and trend of HEX water inlet temperature (Test NAT-3).

the shell side results from the cross sectional area of the shell minus the space taken by 7 tubes. Heat structure H1186 is used to model wall of the inner pipe P186. Both right and left boundary

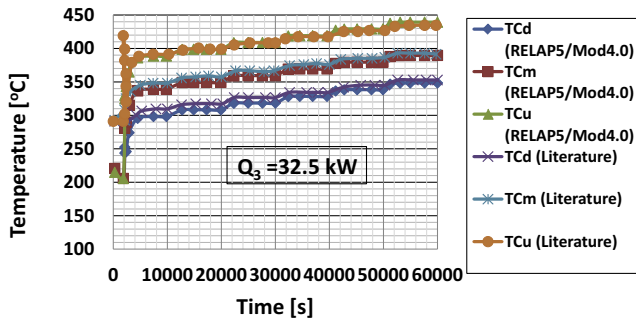


Fig. 10. Comparison of pins surface temperatures (Test NAT-3).

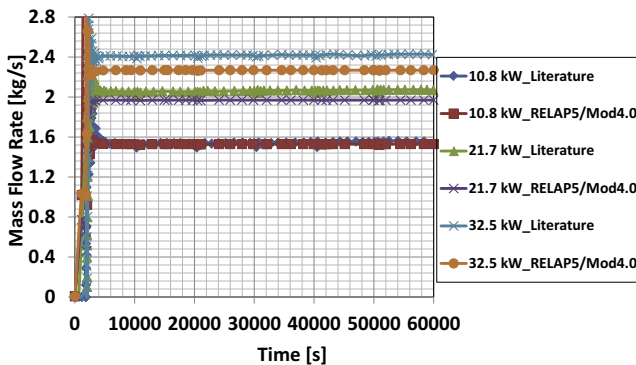


Fig. 11. Mass flow rate of LBE for NAT-1, NAT-2 and NAT-3.

conditions are considered convective (Kleefeldt, 2003; Kumari et al., 2011; Kumari and Khanna, 2014, 2016).

3.3. Results and discussion

All simulations have been performed at three power levels ($Q_1 = 10.8$ kW (NAT-1), $Q_2 = 21.7$ kW (NAT-2) & $Q_3 = 32.5$ kW (NAT-3)) under the natural circulation conditions. Heat section (HS) and heat exchanger (HEX) thermal centre distance has been set at 6.25 m. The effect of heat exchanger water inlet temperature has been studied by 10°C stepwise increase from 120°C to 170°C at the interval of 10,000 s. All simulations have been done for 60,000 s. Power and water supply are kept switched off for 2000 s during startup. Fig. 5 represents effect of inlet water temperature on the inlet and outlet LBE temperature in the HS for NAT-1. Temperatures of LBE reach steady state at around 5000 s.

Table 4

Comparison of temperature difference and HTC of LBE.

NAT		Inlet water temperature (°C)						RMSD
		120	130	140	150	160	170	
Temperature difference [°C] in HS								
1	HS	44 (48)	44(48)	44 (48)	44 (48)	44 (48)	44 (48)	4.0
2	HS	69 (71)	69(71)	69 (71)	69 (71)	69 (71)	69 (71)	2.0
3	HS	90(92)	90(92)	91(92)	91(92)	91(92)	91(92)	1.4
Heat transfer coefficients [W/(m² K)] for HS (bundle)								
1	HS	17,487 (17,458)	17,478(17,682)	17,468 (17,915)	17,459 (18,145)	17,450 (18,373)	17,441 (18,598)	696
2	HS	19,774(19,756)	19,763(19,966)	19,753 (20,174)	19,743 (20,380)	19,732 (20,584)	19,722 (20,786)	643
3	HS	21,956 (21,926)	21,945(22,128)	21,933 (22,312)	21,922 (22,508)	21,910 (22,703)	21,899 (22,866)	590
Heat transfer coefficients [W/(m² K)] of LBE in HEX								
1	HEX (LBE)	955 (949)	955 (961)	954 (975)	953 (988)	952 (1001)	951 (1014)	37
2	HEX (LBE)	1075 (1063)	1074 (1075)	1073 (1087)	1072 (1099)	1071 (1110)	1071 (1122)	29
3	HEX (LBE)	1179 (1182)	1178 (1193)	1177 (1204)	1176 (1215)	1175 (1227)	1174 (1236)	39

Note: Parenthesis represents Barone's results.

It is clear from the figure that when water inlet temperature is increased in the heat exchanger LBE temperature also increases stepwise. Inlet (223°C) and outlet (268°C) temperatures of LBE obtained from simulation of the present work match with the Barone's values of 223°C and 269°C within a difference of 1°C . Fig. 6 represents the pins surface temperatures at three locations along the HS: first volume (TCd at 50 mm), sixth volume (TCm at 300 mm) and twelfth volume (TCu at 600 mm). The profiles of pins surface temperatures match with the Barone's profiles within 5% of relative error. The minimum temperature observed for LBE is 175°C during NAT-1 test ($Q_1 = 10.8$ kW) at the water inlet temperature 120°C which is still higher than the LBE freezing point (124°C) (Barone, 2012).

Fig. 7 depicts the effect of increase in water inlet temperatures on inlet and outlet temperatures of LBE at heat section HS for NAT-2. Inlet (276°C) and outlet (346°C) temperatures of LBE obtained from the simulation of present work match with Barone's values of 272°C and 343°C within a difference of 4°C . Pin surface temperatures at the thermocouples locations TCd, TCm and TCu along the active length of the heat source for NAT-2 have been compared in Fig. 8. Pin surface temperature at TCd (292°K), TCm (323°K) and TCu (361°K) agrees with the literature (285°K , 316°K and 347°K) within 4% of relative error.

Fig. 9 shows effect of water inlet temperature on the inlet and outlet temperatures of LBE at heat section HS with respect to water inlet temperature for NAT-3. Inlet (326°C) and outlet (417°C) temperatures of LBE of the present work match with the Barone's values of 335°C and 427°C within a difference of 10°C . Fig. 10 represents comparison of the pin surface temperature at three predefined locations for NAT-3. Pin surface temperature at TCd (348°K), TCm (390°K) and TCu (439°K) agrees with the literature (353°K , 391°K and 435°K) within 2% of relative error.

Fig. 11 represents mass flow rate of LBE for all the three cases. Barone's LBE mass flow rate slightly increases with the inlet water temperature (Barone, 2012) while the mass flow rate obtained from RELAP5/Mod4.0 simulation remains constant with respect to water inlet temperature. RELAP5/Mod4.0 values match with literature values within a difference of 0.15 kg/s in NAT3.

Temperature difference across HS and heat transfer coefficients of LBE & water across HS and HEX are compared in Table 4. All the results and the corresponding root mean square deviation (rmsd) have been summarized for the three powers 10.8 kW, 21.7 kW and 32.5 kW at different water inlet temperatures. The values reported in parenthesis are from Barone's paper.

Temperature differences across HS increases with the increase of power level. However it remains almost constant with respect to water inlet temperature for all three cases. Temperature

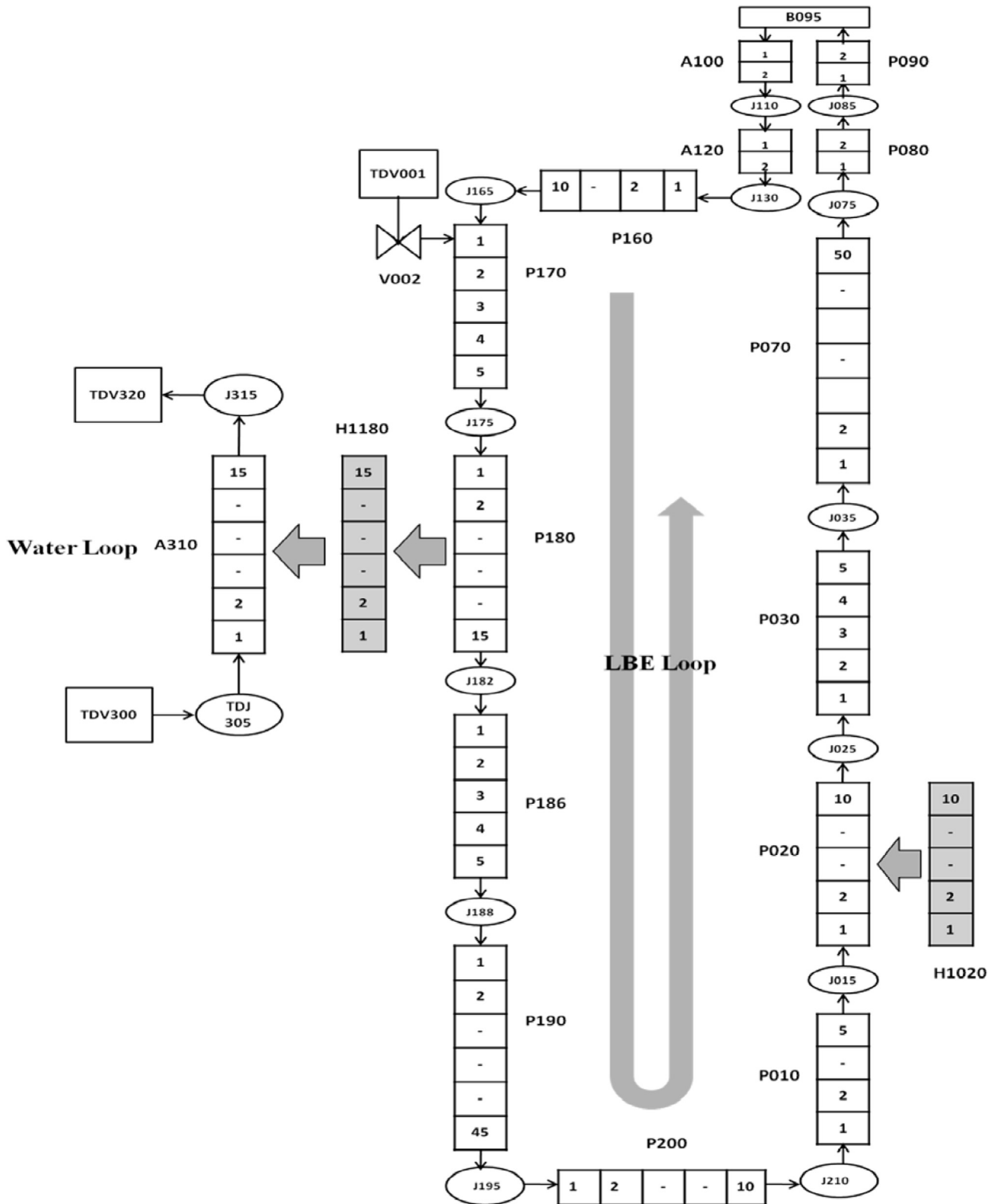


Fig. 12. NACIE facility nodalization (adapted from Tarantino et al., 2010).

difference of this work differs from the literature (Barone, 2012) values within rmsd of 4.0 °C. Heat transfer coefficients of present work decreases with the increase of inlet water temperature while Barone's results follows the opposite trend. However it increases with the increase of power level. Heat transfer coefficients of LBE

obtained in the present work match with Barone's (Barone, 2012) values within rmsd of 696 W/m²K. In the present work LBE specific heat capacity at constant pressure decreases with the increase of water inlet temperature and mass flow rate is observed to be constant. However, in the Barone's paper constant value of

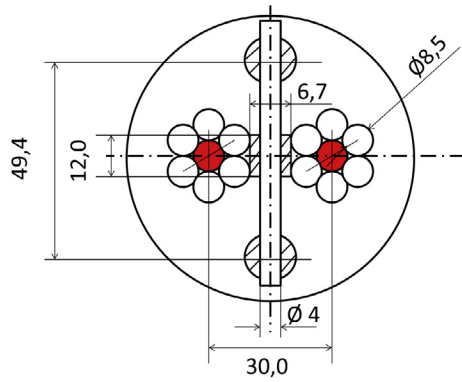


Fig. 13. Cross sectional view of the NACIE bundle (adapted from Tarantino et al., 2010).

Table 5
NACIE bundle details.

Rod diameter	8.2 mm
Active length	850 mm
Number of active pin	2
Thermal power	22.5 kW

Table 6
NACIE heat exchanger details.

	Inner pipe	Middle pipe	Outer pipe
Inner diameter (mm)	62.68	84.9	102.3
Outer diameter (mm)	73.0	88.9	114.3
Thickness (mm)	5.16	2.0	6.02
Length (mm)	1500	1500	1500
Material	AISI 304	AISI 304	AISI 304

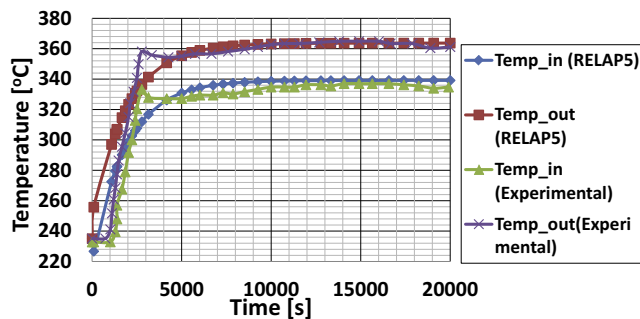


Fig. 14. Comparison of Inlet and outlet LBE temperatures along HS (test no 21).

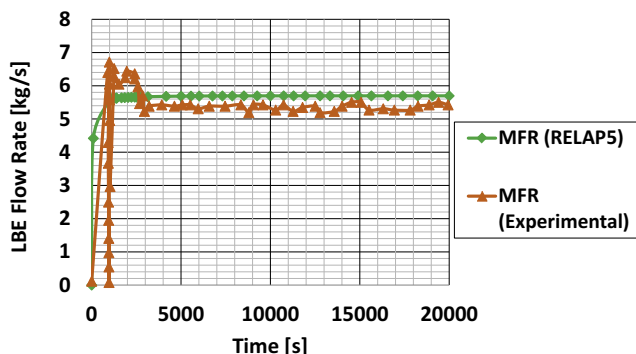


Fig. 15. Comparison of mass flow rate of LBE (test no 21).

$c_p = 147 \text{ J/(kg K)}$ is taken but the mass flow rate is in increasing order resulting in the increase in heat transfer coefficient.

4. Validation of RELAP5/Mod4.0

The NACIE facility by Tarantino et al. (2010) has been considered as the basis for the validation of RELAP5/Mod4.0 code. The dimension of NACIE loop is similar described in the benchmarking section. The difference between the thermal centre of the heat section and centre of the heat exchanger is set to 4.5 m. Test number 21 at power level 22.5 kW of Tarantino et al. (2010) has been considered for the validation of RELAP5/Mod4.0 code. Initial conditions of temperature and pressure are taken as 350 °C and $1.0 \times 10^5 \text{ Pa}$ respectively. Water in the heat exchanger is circulated at low pressure $1.5 \times 10^5 \text{ Pa}$ and at 65 °C temperature. Mass flow rate imposed on water loop is 0.55 kg/s. The RELAP5 nodalization diagram for the NACIE facility is shown in Fig. 12. The model contains several hydrodynamic volumes, single junctions and heat structures. Two coolant loops have been modeled. One loop, called the primary loop, represents the LBE flowing in the counter clockwise direction. The other loop, called the secondary loop, represents the water flowing into the HEX in the counter clockwise direction (Tarantino, 2010; Tarantino et al., 2008, 2010, 2011).

4.1. Boundary conditions

4.1.1. Heat section

Heat section consists of 2 electric rods and two dummy rods to support the fuel bundle. All the four rods are of same outer diameter 8.2 mm. The code has been validated by activating only one fuel rod. A grid spacer as shown in Fig. 13 is installed in the middle of the active length. The dimensions of the electric rods are listed in Table 5. The fuel rod has been simulated by the heat structure H1020 connected to the hydrodynamic component P020. The heat structure is used to calculate the heat across the wall of the hydrodynamic component.

4.1.2. Heat exchanger (HEX)

Tube in tube type heat exchanger is used for cooling the NACIE loop. The heat exchanger consists of three coaxial pipes. LBE flows through the internal pipe in the downward direction while water flow counter currently through the annular region between the middle and outer pipes. Dimensions of the heat exchanger are reported in Table 6. The gap between inner and middle pipes is filled with stainless steel powder of thermal conductivity 3 W/m K (Tarantino et al., 2010, 2011). The pipe P180 representing the internal pipe of HEX and the corresponding heat structure H1180 are divided into 15 volumes of cell length 0.1 m. Volume flow area of the annulus A310 is simply calculated by using inside diameter of 0.1023 m. Wall of internal pipe P180 and the gap between the internal and middle pipes are modeled using heat structure H1180 (Tarantino et al., 2010, 2011).

4.2. Results and discussion

Simulation has been performed by supplying 22.5 kW power using electrical heater under the natural circulation condition for 20,000 s. Gas injection has not been modeled in this paper. Driving force for circulating LBE in the primary loop results from thermal buoyancy. One case (test No. 21) results of the literature (Tarantino et al., 2010) has been considered in this paper for validation of RELAP5/Mod4.0. The system temperature behaviors with time for heat section is depicted in Fig. 14. Fig. 14 represents comparison of inlet and outlet temperature of the heat section against the literature. The system takes approximately 5000 s to attain the

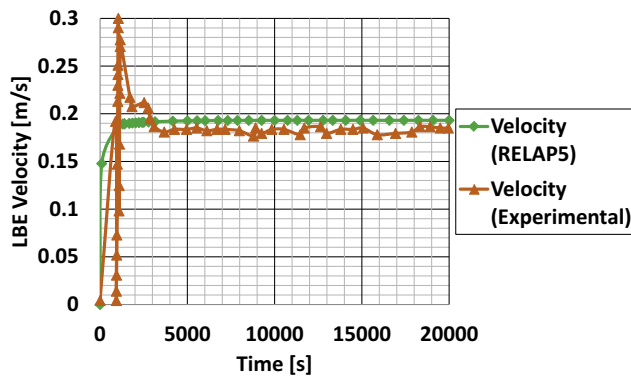


Fig. 16. Comparison of LBE velocity (test no 21).

steady state condition. Outlet temperature 364 °C obtained from the simulation matches with the experimental value 365 °C with the difference of 1 °C. Similarly the difference observed between the simulated 339 °C and experimental 338 °C inlet temperature is also 1 °C.

Mass flow rate of LBE in the primary loop is shown in Fig. 15. Experimental value (~6.5 kg/s) of LBE mass flow rate is observed to be higher up to 3000 s. At 3000 s, the mass flow rate goes down rapidly and attain a steady state value of ~5.5 kg/s. However, the mass flow rate of LBE obtained in RELAP5 simulation reached a steady state value ~5.7 kg/s at 2000 s. The simulated flow rate of LBE match with the experimental value with difference of 0.2 kg/s (Tarantino et al., 2010, 2011).

The LBE velocity time trends obtained from RELAP5 simulation and experimental results are compared in Fig. 16. RELAP5/Mod4.0 overestimates the LBE velocity with respect to the experimental one. Simulated LBE velocity (0.19 m/s) agrees well with the experimental values (0.18 m/s) with the discrepancy of 0.01 m/s (Tarantino et al., 2010, 2011).

5. Conclusions

RELAP5/Mod4.0 code has been benchmarked against Barone's paper which simulates the NACIE facility using RELAP5/Mod3.3. Three case studies NAT-1 (for 10.8 kW), NAT-2 (for 21.7 kW) and NAT-3 (for 32.5 kW) reported in the literature have been benchmarked in this work. Effect of heat exchanger water inlet temperature has been studied by 10 °C stepwise increase from 120 °C to 170 °C at the interval 10,000 s. Mass flow rates of LBE match with their values within 6% of relative error; LBE heat transfer coefficients in HS and HEX match with literature values within 4% relative error; inlet and outlet temperatures and temperature difference of LBE in heat section match with the Barone's values within a difference of 10 °C. Pins surface temperatures obtained in this work differ from the literature values within 5 °C.

RELAP5/Mod4.0 code has been validated against the experimental results of Tarantino et al. for the NACIE facility. Simulation has been carried out at 22.5 kW power for 20,000 s under the natural circulation condition. One case study test No. 21 reported in

the literature has been considered in this work. Inlet and outlet temperatures obtained from the simulation agree well with experimental value with difference of 1 °C. The simulated flow rate of LBE match with the experimental value with difference of 1 kg/s. Simulated LBE velocity (0.19 m/s) agrees well with the experimental values (0.18) with difference of 0.01 m/s.

The preliminary validation of LBE properties in the modified RELAP5/Mod4.0 code has been performed in the present work. The modified code gives the acceptable results, but future work is required to improve the implements of LBE.

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